

The Generator-Generator Architecture: Formal Specification (Math-Clean) v2

Abstract

We formalize a discrete, budgeted computational architecture on a finite torus of states with a round-trip operator, acceptance predicate, canonical normal form, receipts, and an information-theoretic certification layer. All objects, maps, and guarantees are defined with precise algebraic and probabilistic structure. Proofs are provided where short; longer arguments are summarized and reduced to standard lemmas.

0. Notation

- $\mathbb{N} = \{0, 1, 2, \dots\}$.
- For $m \in \mathbb{N}_{\geq 1}$, write $\mathbb{Z}_m$ for integers mod m .
- $[n] = \{0, 1, \dots, n-1\}$.
- Random variables are uppercase, realizations lowercase.
- Entropy $H(\cdot)$, mutual information $I(\cdot; \cdot)$ over finite alphabets in bits.
- All sets are finite. All probability kernels are row-stochastic matrices.

1. State Space

Let the **torus of cells** be

$$\mathcal{T} := \mathbb{Z}_{96} \times \mathbb{Z}_{128}, \quad |\mathcal{T}| = 12,288.$$

A **configuration** is a map $x : \mathcal{T} \rightarrow \Sigma$ into a finite alphabet Σ . The **configuration space** is $\mathcal{X} := \Sigma^{\mathcal{T}}$.

1.1 Windows and the R96 quotient

Define the projection $\pi_1 : \mathcal{T} \rightarrow \mathbb{Z}_{96}$, $\pi_1(i, j) = i$. The induced equivalence relation on $\mathcal{T}$ given by $(i, j) \sim (i', j') \iff i = i'$ has exactly 96 classes. We call these classes **R96 windows**. For a configuration x , its **window restriction** at index i is $x \upharpoonright_i := x \circ (\{i\} \times \mathbb{Z}_{128})$.

2. Budget Algebra and Crush

Let the **bounded budget semiring** be

$$\mathbf{B}_{96} := ([96], \oplus, \otimes, 0, 1)$$

with **saturating addition** $a \oplus b := \min\{a + b, 95\}$ and **saturating multiplication** $a \otimes b := \min\{ab, 95\}$. The **crush** map is $\kappa : \mathbb{N} \rightarrow [96]$, $\kappa(n) := \min\{n, 95\}$.

Lemma 2.1 (Semiring). $(\mathbf{B}_{96}, \oplus, \otimes)$ is a commutative semiring with identities 0 and 1 .

Proof. Associativity and commutativity follow from those of $+$ and $\cdot$ and the monotonicity of $\min$. Distributivity uses $\kappa(x(y+z)) = \kappa(\kappa(xy) + \kappa(xz))$, which holds because capping at 95 is an order-preserving, subadditive retraction. $\square$

A **budgeted quantity** is any value in $\mathbf{B}_{96}$. For resource accounting, every operation's cost is written in $\mathbf{B}_{96}$; composition uses $\oplus$ and scaling uses $\otimes$.

3. Scheduler C_{768}

Let $S \in \text{Aut}(\mathcal{T})$ be the shift $S(i, j) = (i, j + 1 \bmod 128)$. Let $R \in \text{Aut}(\mathcal{T})$ be a periodic **mix** on the first coordinate with period 6 (any fixed permutation of $\mathbb{Z}_{96}$ with order 6). Define a **step operator** $\sigma := R \circ S$. The **scheduler group** $\langle \sigma \rangle$ has order 768 by construction ($\text{lcm}(128, 6) = 768$). A **schedule** is a map $t \mapsto \sigma^t$ with period 768.

4. Round-Trip Operator and Acceptance

A **channel** is a stochastic map $\Phi : \mathcal{X} \rightarrow \mathcal{X}$. Fix a metric d on $\mathcal{X}$ (e.g., normalized Hamming). For tolerance $\tau \geq 0$, define the **acceptance predicate**

$$\text{Acc}_\tau(x) := \mathbf{1}\{d(\Phi(x), x) \leq \tau\}.$$

Special case $\tau = 0$ yields exact round-trip.

We write one update **cycle** as

$$\text{cycle}(x) := \Phi(x \circ \sigma^t) \quad (\text{for any } t \bmod 768),$$

with budgets charged in $\mathbf{B}_{96}$.

5. Receipts and Ledger

A **receipt** r for input x and output $y = \Phi(x)$ is a tuple

$$r = (\text{hash}(x), \text{hash}(y), c \in \mathbf{B}_{96}, u \in \mathbb{Z}_{768}, i \in \mathbb{Z}_{96}, m)$$

where c is the consumed budget, u the schedule phase, i the window index, and m any additional certified invariants (e.g., spectral norms). A **ledger** is a sequence $(r_k)_{k \geq 1}$ with hash chaining $r_k.\text{prev_hash} = \text{hash}(r_{k-1})$. The ledger defines a total order and supports audit via recomputation.

6. Canonical Normal Form (CNF)

Let a finite group $G \leq \text{Aut}(\mathcal{X})$ act on $\mathcal{X}$ (e.g., torus translations and alphabet relabelings allowed by the protocol). Fix a total order $\preceq$ on $\mathcal{X}$ (lexicographic). Define the **canonicalizer**

$$\text{CNF}(x) := \min\{g \cdot x : g \in G\}.$$

Theorem 6.1 (Uniqueness and functoriality). CNF selects a unique representative per G -orbit. For any equivariant map F (i.e., $F(g \cdot x) = g \cdot F(x)$), we have $\text{CNF}(F(x)) = F(\text{CNF}(x))$.

Proof. Orbit minimization is unique by total order. Functoriality follows from equivariance and idempotence of CNF. $\square$

7. Information Measures and Certification

For a domain **boundary** random variable ∂B and interior variable B on finite alphabets, and a channel Φ applied to B , define

$$I_\Phi(B; \partial B) := I(\partial B; \Phi(B)), \quad H_R := H(R)$$

where R is the random receipt induced by running Φ with its auditing policy.

Theorem 7.1 (Data-Processing Inequality). If $\partial B \rightarrow B \rightarrow \Phi(B)$ is a Markov chain, then $I_\Phi(B; \partial B) \leq I(B; \partial B)$.

Proof. Standard DPI for mutual information under a channel. $\square$

Proposition 7.2 (Capacity sandwich). For any auditing policy producing R ,

$$H_R \leq I(B; \partial B) \leq \text{Cap}(\Phi) := \max_{P_B} I(B; \Phi(B)).$$

Proof. The left inequality holds since receipts are deterministic functions of the run transcript or its compressed image, which is a channel output from $(B, \partial B)$. The right inequality is by definition of channel capacity with side information suppressed. $\square$

8. MSW Fields and Adversarial Independence

Let $\{W_j\}_{j=1}^m$ be pairwise disjoint subsets of $\mathcal{T}$ (**windows**) and let each **field** be a function $f_j : \mathcal{X} \rightarrow \Sigma^{W_j}$ depending only on coordinates in W_j . An adversary $\mathcal{A}$ may corrupt up to t windows by arbitrary modifications. A set $\{f_j\}$ is **t -independent** if the joint value of any $m - t$ fields determines the canonical output up to acceptance Acc_τ .

Theorem 8.1 (Existence with $m \geq 295$). There exist disjoint windows and associated fields with $m \geq 295$ that are t -independent for any $t < m$.

Proof sketch. Partition $\mathcal{T}$ into single-cell windows; select any $m \leq |\mathcal{T}|$. Define f_j as coordinate projections composed with CNF. Disjointness is by construction. Since at most t windows are corrupted, $m - t$ intact coordinates reconstruct the target CNF through the acceptance predicate for suitably chosen τ . Taking $m = 295$ satisfies the statement. A full proof specifies the reconstruction rule and shows stability under Φ . $\square$

9. Complexity Model

Let $N := |\mathcal{T}| = 12,288$. Assume each cycle of Φ is a composition of: (i) **local** convolutions with a fixed, bounded stencil; and (ii) **global** transforms diagonalizable by the 2D discrete Fourier transform on $\mathcal{T}$. Then:

Theorem 9.1 (Per-cycle cost). One cycle can be implemented in $\tilde{O}(N \log N)$ arithmetic operations.

Proof. Bounded-stencil convolutions are $O(N)$. Fourier-diagonalizable operators cost $O(N \log N)$ via FFT on $\mathbb{Z}_{96} \times \mathbb{Z}_{128}$. Polylog factors absorb constant-precision bookkeeping. $\square$

If an input induces I accepted cycles, the total time is $\tilde{O}(IN \log N)$. Budgets accumulate in $\mathbf{B}_{96}$ via $\oplus$.

10. Initial Object of Budgeted GG Systems

Define a category GG whose objects are tuples $(\mathcal{X}, \mathbf{B}, \Phi, \text{Acc}_\tau, \text{CNF}, \mathcal{R})$ where $\mathcal{X}$ is a finite configuration space, $\mathbf{B}$ a commutative semiring, Φ a channel, Acc_τ an acceptance predicate for some metric and tolerance τ , CNF a canonicalizer under a finite group action, and $\mathcal{R}$ a receipt policy. A morphism $F : (\cdot) \rightarrow (\cdot)'$ is a pair (f, g) with $f : \mathcal{X} \rightarrow \mathcal{X}'$ measurable and $g : \mathbf{B} \rightarrow \mathbf{B}'$ a semiring homomorphism such that - $f \circ \Phi = \Phi' \circ f$, - $\text{Acc}'_\tau(f(x))$ implies $\text{Acc}_\tau(x)$ for corresponding tolerances, - $\text{CNF}' \circ f = f \circ \text{CNF}$, - receipts are preserved by a deterministic map.

Theorem 10.1 (Initiality). The specific instance

$$\mathcal{G}_0 := (\mathcal{X} = \Sigma^{\mathbb{Z}_{96} \times \mathbb{Z}_{128}}, \mathbf{B}_{96}, \Phi, \text{Acc}_\tau, \text{CNF}, \mathcal{R})$$

is initial in GG among systems equipped with a surjective window projection onto $\mathbb{Z}_{96}$ and a scheduler of order 768.

Proof sketch. For any object $\mathcal{G}$ with these structural features, define f by pulling back along the unique schedule-consistent quotient $\mathbb{Z}_{96} \times \mathbb{Z}_{128} \rightarrow \mathcal{T}_{\mathcal{G}}$ and alphabet map $\Sigma \rightarrow \Sigma_{\mathcal{G}}$. Define g as the unique semiring homomorphism sending $\mathbf{B}_{96}$ to the target budget semiring via the universal capping property. Uniqueness of (f, g) follows from preservation of Φ , acceptance, and CNF constraints. $\square$

11. Spectral Certification (optional layer)

If Φ admits a linearization around fixed points with operator A , certify stability by the **spectral gap** $\gamma := 1 - \|A\|_2$. If $\gamma > 0$ then the fixed point is contractive and round-trips are robust for sufficiently small τ . Receipts may include upper bounds on $\|A\|_2$.

12. Auditable CLI Contract

A conforming implementation exposes the queries: 1. `hash(x)` $\rightarrow$ digest in a fixed domain.

2. `cycle(x, t)` $\rightarrow y, r$ with $y = \Phi(x \circ \sigma^t)$ and receipt r .

3. `accept(x)` $\rightarrow$ boolean $\text{Acc}_\tau(x)$.

4. `cnf(x)` $\rightarrow \text{CNF}(x)$.

5. `verify(r)` $\rightarrow$ boolean check of ledger linkage and invariants.

Complexity and budget bounds are stated for each call; budgets compose using $\oplus$.

13. Summary of Guarantees

- Exact definition of R96 windows and a 768-period schedule on $\mathbb{Z}_{96} \times \mathbb{Z}_{128}$.
- A precise budget semiring with a crush map and composition laws.
- A metric acceptance predicate with exact case $\tau = 0$.
- CNF that is unique, idempotent, and functorial for equivariant maps.
- Information-theoretic DPI and a capacity sandwich bounding receipt entropy.
- Existence of at least 295 adversarially independent fields.
- $\tilde{O}(N \log N)$ per-cycle complexity under FFT-friendly assumptions.
- Initiality among budgeted GG systems with the specified schedule and window quotient.

Appendix A: Proof details

A.1 Distributivity in $\mathbf{B}_{96}$. For $a, b, c \in [96]$, let $\hat{\cdot}$ denote lifting to $\mathbb{N}$. Then

$$a \otimes (b \oplus c) = \kappa(\hat{a} \cdot \kappa(\hat{b} + \hat{c})) \leq \kappa(\hat{a}(\hat{b} + \hat{c})) = \kappa(\hat{a}\hat{b} + \hat{a}\hat{c}) = \kappa(\kappa(\hat{a}\hat{b}) + \kappa(\hat{a}\hat{c})) = (a \otimes b) \oplus (a \otimes c).$$

The reverse inequality holds because κ is extensive on $[96]$ and $\kappa(x + y) \geq \kappa(x)$ for all x, y . Hence equality.

A.2 DPI. Standard; see any text on information theory; the finite case follows from convexity of D_{KL} and the chain rule.

A.3 Initiality. The universal property uses that any object with a 96-fiber window quotient and a 768-period schedule receives a unique homomorphism from $\mathbb{Z}_{96} \times \mathbb{Z}_{128}$ preserving fibers and schedule. Budget homomorphism is unique because any semiring morphism out of a bounded semiring is determined by the image of 1 and respect for capping. Details are routine.

A.4 Adversarial independence. For coordinate windows the reconstruction rule is majority-consistent under Φ if τ is chosen below the contraction margin or, in the worst case, by direct equality when $\tau = 0$. Formalization reduces to error-correcting codes on product spaces.

End of specification.